

Surplus Food Redistribution Network

Mlo Mtaani | Team Ember Alliance | UN SDG Goal 2: Zero Hunger

Team Members: Hellen Mucheru (Product Lead/Strategy), Idris Muhammad (Backend/Automation), Farida Abdul Ganew (Research/QA)

Executive Summary

Mlo Mtaani is a surplus food redistribution network designed to reduce food waste and improve meal access for food-insecure communities in Kenya. It addresses a coordination gap in which small and mid-size grocery vendors discard edible perishables because they lack a fast, local system for routing surplus to nearby pantries and volunteers.

The platform connects vendors, community pantries, and volunteers through a lightweight three-sided model. Vendors upload surplus listings in under 60 seconds, pantries receive ranked matches based on proximity, expiry urgency, and need, and volunteers claim pickup tasks with navigation and delivery confirmation.

The five-week implementation plan covers discovery, design, core development, testing, and presentation. Python will support the matching logic, Firebase will provide database and authentication services, and mapping tools will handle geolocation. Success will be measured through listing time, match speed, delivery completion, kilograms redirected, and estimated meals served.

1. Research and Data

1.1 Problem Statement and Local Context

Small and mid-size grocery vendors routinely discard perishable food because they lack a fast, hyper-local method to route surplus to nearby pantries before it spoils.

- **Nairobi Household Food Waste:** Estimated at 91 to 100 kg per person annually (UN-Habitat/JICA via UNEP).
- **National Impact:** Up to 40% of food produced in Kenya is lost or wasted annually, valued at ~KES 72 billion (~US\$578M).
- **Hunger Gap:** Roughly 33% of Kenyans are food-poor, and 25% face severe food insecurity. Kenya ranks 90th out of 125 on the Global Hunger Index ("serious" level).
- **Core Insight:** The primary barrier is not a lack of food or demand, but the missing hyper-local coordination layer connecting vendors, pantries, and volunteers.

1.2 Competitive Landscape and Market Gap

Existing solutions present distinct limitations in hyper-local perishable food delivery:

- **Too Good To Go:** Consumer-focused retail discount channel, not targeted at food-insecure households.
- **OLIO:** Peer-to-peer manual browsing with no automated distance or expiry matching.
- **Food Rescue US / FareShare / Feeding America:** Rely heavily on manual volunteer opt-ins or central warehousing, which slows down the redistribution of fast-spoiling perishables.
- **Ember Alliance Gap:** Implements an automated, explainable matching engine that ranks listings by distance, expiry urgency, and pantry need.

2. Solution Development

2.1 Product Overview

Mlo Mtaani ("a meal in the neighbourhood") is a lightweight, three-sided platform connecting grocery vendors, community pantries, and local volunteers.

2.2 Core Architecture and User Flows

1. **Vendor Application:** Simple intake form allowing vendors to list surplus items, quantities, expiry windows, and pickup locations in under 60 seconds.
2. **Automated Matching Engine:** A rule-based weighted-scoring formula computes match priority:
$$\text{match_score} = (w1 \times \text{proximity_score}) + (w2 \times \text{urgency_score}) + (w3 \times \text{pantry_need_score})$$
3. **Pantry Application:** Presents a ranked shortlist allowing one-tap reservations.
4. **Volunteer Application:** Lists claimable pickup tasks with direct navigation links and one-tap delivery confirmations.
5. **Admin Dashboard:** Tracks impact metrics including kilograms redirected, estimated meals served, matches completed, and estimated CO2 avoided.

2.3 Technology Stack

- **Backend:** Python (Matching Engine, logic).
- **Database & Auth:** Firebase (Firestore, Firebase Auth, Hosting).
- **Mapping/Location:** Google Maps API (Geocoding & Proximity calculations).
- **Optional AI Layer (Phase 2+):** Gemini API for natural-language impact summaries and demand prediction.

3. Implementation Plan

3.1 Five-Week Execution Pipeline (July 15 – August 14, 2026)

- **Week 1 (Discover and Set Up):** Define user personas, establish GitHub repository, conduct 5–8 local stakeholder interviews with vendors, pantries, and volunteers.
- **Week 2 (Design):** Finalize ER diagrams, API surface, user journey maps, wireframes, and Business Model Canvas.
- **Week 3 (Build Core):** Implement core matching logic in Python, Firebase integration, and initial end-to-end user flows.
- **Week 4 (Enhance & Test):** Execute testing plan (unit, integration, usability, performance), polish UI, and integrate the admin impact dashboard.
- **Week 5 (Finish and Present):** Code freeze, demo video recording, and submission package completion, targeted for Thursday completion to respect Jumu'ah.

3.2 Key Performance Indicators

- **Listing Time:** Under 60 seconds per vendor upload.
- **Match Speed:** Under a few minutes from listing creation to pantry shortlist generation.
- **Delivery Success Rate:** Maximum completion rate of accepted pickups.

3.3 Risk Management and Mitigation Strategies

Risk	Severity	Mitigation Strategy
Low Vendor Adoption	High	Keep onboarding under 1 minute; begin pilot with 2–3 pre-validated local vendors.
Food Safety Liability	High	Include clear expiry and handling disclaimers on all listings; maintain transparent quality messaging.
Volunteer No-shows	Medium	Maintain a backup volunteer queue and automated push notifications.
Connectivity Gaps	Medium	Implement local offline data caching and an SMS/WhatsApp fallback intake.
Data Privacy	High	Limit data collection to operational necessities; comply with Kenya’s Data Protection Act (2019).

4. Conclusion

Mlo Mtaani offers a practical response to food waste and food insecurity by linking vendors, pantries, and volunteers through a simple automated matching system. It targets the coordination gap that prevents edible surplus from reaching people who need it most.

The solution is feasible within the project timeline because it focuses on a lightweight minimum viable product, clear user flows, and measurable indicators. Familiar tools such as Python, Firebase, and mapping services support rapid development and future expansion.

With careful piloting, stakeholder engagement, and risk management, Mlo Mtaani can support UN SDG Goal 2: Zero Hunger while creating a scalable model for community-based food rescue in Kenya.

5. References

Concern Worldwide and Welthungerhilfe. (2024). *2024 Global Hunger Index*.

Kenya Law. (2019). *Data Protection Act, Cap. 411C*.

United Nations Environment Programme. (2024). *Food Waste Index Report 2024*.

United Nations. (2015). *Sustainable Development Goal 2: Zero Hunger*.

World Resources Institute Africa. (2025). *Turning the tap on food loss and waste in Kenya*.